

What parameter-wise symmetry attribution can and cannot identify

Companion notes to the NeurReps retarget. Every proposition below is stated for the setting of the SCML draft (§2.2, §2.4) and is verified numerically by a script in this repo; the verification residual is quoted with each one.

All numbers below are full scale: the Mexican-hat ASRNN, $P = 4514$ (2221 of which can affect the force at all — K_θ cannot), trained 8000 steps to trajectory loss 6.7×10^{-5} , equivariance residual 9.1×10^{-3} and $\|\delta\|/\|f\| = 6.6 \times 10^{-3}$. This is squarely the “network with low equivariance error” regime the claim is stated for. A second, non-augmented model at matched convergence (loss 6.4×10^{-5} , equivariance residual 4.2×10^{-3}) reproduces every qualitative finding.

A note on scale-dependence, because it matters for how these results should be read: the conditioning confound is **much worse after convergence**. On a lightly-trained toy model the column norm explains 15% of the variance of $\log |c_i|$; on the converged full-scale model it explains **91%**. Convergence spreads $\|S_i\|$ over eight orders of magnitude, and that spread is what the attribution ends up reporting.

Throughout: $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^m$, sensitivities $S_i = \partial f_\theta / \partial \theta_i$ evaluated on a probe set and flattened, Jacobian $J \in \mathbb{R}^{N \times P}$ with columns $J_{:,i} = S_i$, truncated SVD $J \approx U_r \Sigma_r V_r^\top$ at relative cutoff τ , and a symmetry generator direction $g = X f_\theta$. Write $d_i = \|S_i\|$, $D = \text{diag}(d)$, and $\alpha = \Sigma_r^{-1} U_r^\top g$, so that the constraint $Jc = P_{T_\theta} g$ is equivalent to $V_r c = \alpha$.

0. The solution set

Proposition 0 (underdetermination). The set $\{c : Jc = P_{T_\theta} g\}$ is an affine subspace of dimension $P - r$.

In these models r is a few tens to a few hundred while P is thousands, so the solution set has dimension in the thousands. *Every* point in it reproduces the identical function-space direction. Selecting one requires minimising something over it, and the data does not say what.

This is the root of everything below: “which parameters realise the symmetry” is not a question about f_θ alone until a selection rule is fixed.

1. The conditioning factor

Proposition 1 (conditioning–alignment factorisation). The minimiser of $\|Dc\|_2$ subject to $V_r c = \alpha$ is

$$c_i = \frac{a_i}{d_i^2}, \quad a_i = \langle V_{:,i}, u \rangle, \quad u = (AA^\top)^{-1} \alpha, \quad A = V_r D^{-1}.$$

Proof. Substituting $\tilde{c} = Dc$, the problem is $\min \|\tilde{c}\|_2$ subject to $A\tilde{c} = \alpha$, whose solution is $\tilde{c} = A^\top (AA^\top)^{-1} \alpha$. Then $\tilde{c}_i = (V_{:,i} \cdot u)/d_i$ and $c_i = \tilde{c}_i/d_i$. $\square$

The factor $d_i^{-2} = \|S_i\|^{-2}$ is **target-independent**: identical for the true generator, for a random target, for anything. Only a_i ever knew about the symmetry. So a spread in $|c_i|$ is evidence about symmetry only to the extent that a_i , not d_i , produces it.

Verified: residual 6.4×10^{-16} at full scale (`causal_symmetry_control.py`; the check must replicate the solve’s zeroing of dead columns, or it compares against a different estimator).

Measured at full scale. $\|S_i\|$ spans **8.0 decades** over live columns. Regressing $\log |c_i|$ on $\log \|S_i\|$:

quantity	slope	R^2
$\log c_i $	−0.99	0.906
$\log a_i $ (alignment)	+1.01	0.910
$\log \tilde{a}_i $	−0.000	0.000

So 91% of the variance of the attribution ranking is predicted by a target-blind quantity. (The variance decomposition into alignment/conditioning shares is *not* a useful summary here — the cross term is −374%, because a_i and $\|S_i\|$ are themselves strongly correlated. R^2 and the slope are the honest statistics.)

Correction to §2 of the review notes. $\tilde{a}_i$ is measured here to be *exactly* conditioning-neutral (slope −0.000, $R^2 = 0.000$), not partially so. It fully fixes the conditioning confound. It does not fix gauge dependence (Prop. 6), which is a different problem.

The sparsity claim does not survive. Share of total attribution carried by the top 10% of parameters:

	share
$ c_i $, true generator	99.9%
$ c_i $, matched-construction null	99.9%
$1/\ S_i\ ^2$, conditioning alone	100.0%

The draft’s headline structure — a small subset carrying essentially all the attribution — is reproduced exactly by a random target and by the conditioning factor on its own. And the ranking is largely target-independent: $\text{Spearman}(|c(g)|, |c(\text{null})|) = 0.75\text{--}0.92$ across five null draws.

Corollary 1.1 (exact null calibration). For two targets g, h solved on the same J with the same retained subspace, d_i^{-2} cancels identically:

$$r_i = \frac{|c_i(g)|}{|c_i(h)|} = \frac{|a_i(g)|}{|a_i(h)|}.$$

Taking h over a family of matched-construction nulls and using the geometric mean gives a **provably conditioning-free attribution** — no regression, no matching argument, no distributional assumption.

Verified: cancellation to 4×10^{-16} .

2. Gauge dependence

Let $\Lambda = \text{diag}(\lambda)$, $\lambda_i > 0$, and reparametrise $\theta_i \mapsto \lambda_i \theta_i$. This renames coordinates and changes nothing about the learned function: $S_i \mapsto S_i/\lambda_i$, $J \mapsto J\Lambda^{-1}$, and the *true* coefficients must transform covariantly, $c \mapsto \Lambda c$.

Proposition 2 (the draft’s choice is not covariant). The Moore–Penrose selection $c^* = J^+g$, i.e. $D = I$, does not commute with Λ : it minimises $\|c\|_2$, and $\|\Lambda c\|_2$ is a different objective, so the selected point is not Λ times the original.

Measured (full scale, two-decade change of units): relative departure from covariance 0.998 for the draft’s J^+g and 0.954 for the code default at the working cutoff.

Proposition 3 (column normalisation is covariant — as an objective). With D the column norms, $D \mapsto D\Lambda^{-1}$ and $\|Dc\|_2$ is invariant, so the objective is covariant by construction.

Proposition 4 (but the truncation is not). The retained subspace is defined by discarding singular values below a *relative* cutoff. Column scaling changes the singular spectrum of J , so a different subspace is retained and P_{T_θ} itself moves. Covariance therefore fails even for the covariant objective.

Measured at full scale (on columns clear of the floor of Prop. 4b, so this effect is seen alone):

cutoff	rank(J)	rank($J\Lambda^{-1}$)	$\ \Delta P_T g\ /\ g\ $	departure
10^{-3} (working)	22	19	2.0×10^{-1}	1.13
10^{-5}	40	38	1.3×10^{-2}	1.63
10^{-8}	101	86	1.5×10^{-5}	0.61
10^{-12}	336	336	1.9×10^{-15}	2.6×10^{-7}

Covariance returns only at 10^{-12} , where nothing is truncated — precisely the regime the truncation exists to prevent (Σ_r^{-1}) amplifying numerically meaningless directions).

Proposition 4b (an implementation source, worth fixing regardless). The dead-column rule floors column norms at 10^{-8} times the *largest* column. That floor is relative to a quantity that itself changes under Λ , so it does not commute with a change of units. It only bites when the sensitivity spectrum is wide enough for live columns to sit near it — which is exactly what convergence produces. Here 13 live columns lie within $10\times$ of the floor, and their presence alone moves the departure from 2.6×10^{-7} (columns clear of the floor) to 0.978 (all live columns). An absolute-scale-free criterion, or carrying the floor through the reparametrisation, would remove this.

The dilemma. Covariant-but-unstable, or stable-but-gauge-dependent. There is no cutoff at which both hold.

Proposition 5 (no neutral gauge, and the least-bad one). Under $\|D^s c\|_2$ the explicit factor is d_i^{-2s} , so every choice commits to a preference along the conditioning axis:

gauge	$\rho(c_i , \ S_i\)$	prefers
$s = 0$, the draft’s J^+g	+0.918	most sensitive parameters
$s = 1$, column normalisation	−0.883	least sensitive parameters
$s = 1/2$	+0.039	~neutral
elastic net / L_1	+0.117 / +0.102	~neutral

If a min-norm attribution is wanted at all, $s = 1/2$ is the choice to defend.

Consequence. At full scale the solution set has dimension **4492** (rank 22 of 4514). Mean cross-gauge overlap of the top-20 parameters is **0.140**, and the column-normalised default has overlap **0.000 with every one of the seven other gauges** — disjoint sets, from the same network, reproducing the same function-space direction to within the truncation tolerance.

Proposition 6 (the proposed fix is orthogonal to the problem). The review notes’ $\tilde{a}_i = c_i \langle S_i, g \rangle / \|P_T g\|^2$ is invariant under $\theta_i \mapsto \lambda_i \theta_i$, as claimed. But gauge choice is not reparametrisation: different gauges select genuinely different points of the $(P - r)$ -dimensional solution set, and $\tilde{a}$ inherits that ambiguity. *Measured at full scale:* cross-gauge top-20 Jaccard 0.187 for $\tilde{a}_i$ versus 0.140 for $|c_i|$ — no repair. $\tilde{a}_i$ is the right fix for the conditioning confound (Prop. 1) and no fix at all for this one.

Verified: gauge_dependence.py, reparametrisation_covariance.py.

3. Permutation

Proposition 7 (parameter indices are not well-defined targets). For a hidden layer, permuting units and correspondingly permuting the next layer’s columns leaves f_θ pointwise unchanged for any activation. Any quantity offered as “which parameters carry the symmetry” must therefore be permutation-equivariant, so only its orbit — the multiset of values — is well-defined. Individual weight indices are not comparable across training runs and are not intrinsic within one.

The permutation acts on **hidden units** by relabelling, so a unit-level score *is* permutation-equivariant and its top- k multiset is invariant. The hidden unit is therefore the finest granularity at which localisation of a symmetry can be stated; modules are coarser and also safe.

4. What is well posed

Proposition 8 (E_i is invariant). $E_i = \|S_i(h \cdot x) - \rho(h)S_i(x)\|/\|S_i(x)\|$ is built from norms of the same column of two Jacobians, with no optimisation and no truncation, so numerator and denominator both scale by λ_i^{-1} and E_i is exactly invariant.

Measured at full scale: 2.8×10^{-15} , under the same reparametrisation that moves c_i by 0.95–1.00. **E_i is the well-posed half of the framework** — the draft’s Fig. 1b stands where Fig. 1a does not.

Proposition 9 (coarse-graining restores gauge-invariance). Aggregating $|c_i|$ to a permutation-safe granularity recovers most of what the parameter level loses. Mean cross-gauge Spearman:

granularity	cross-gauge ρ
individual parameters	+0.163
hidden units	+0.711
modules	+0.873

This is the constructive form of the localisation claim: **symmetry localisation in parameter space is well posed at module level, partly at hidden-unit level, and not at the level of individual weights.**

Also well posed, because they are function-space quantities and never reference a metric on parameter space:

- $P_{T_\theta}g$ and the representation residual $\|P_{T^\perp}g\|/\|g\|$;
- the order parameters $\|g\|$ and $\|\delta\|$;
- **gauge-free subset selection:** $\min\|c\|_0$ s.t. $Jc \approx g$ — “which k parameters *suffice* to reproduce the symmetry direction” contains no metric on parameter space. L_1 /elastic net are its standard convex relaxations, consistent with those two agreeing (top-20 Jaccard 0.90) while disagreeing with every L_2 gauge;
- causal interventions on a given support.

5. The regime problem

For an exactly equivariant force field the rotational Lie derivative vanishes identically, so $g = Xf_\theta \rightarrow 0$ as $f_\theta \rightarrow \ker X$; so does $\delta = Qf_\theta$. Both are order parameters for symmetry breaking and they vanish *together* (Prop. 1 of the review notes says only that they are orthogonal while doing so).

The attribution is therefore a direction estimated from a vanishing quantity, exactly in the “low equivariance error” regime the claim is stated for. The prediction is that identifiability of the symmetry-specific ranking degrades as equivariance improves, while $|c_i|$ continues to look clean and sparse — because its apparent structure is carried by d_i^{-2} , which neither vanishes nor cares about the symmetry.

Measured (identifiability_vs_equivariance.py, 11 checkpoints, augmented model, equivariance residual falling by a factor of 250 across training):

step	equivariance	$\ g\ /\ f\ $	repr. error	$r - r'$ (unit)
0	1.2×10^0	1.1×10^0	0.001	+0.116
160	5.9×10^{-2}	5.7×10^{-2}	0.017	+0.192

step	equivariance	$\ g\ /\ f\ $	repr. error	$r - r'$ (unit)
1600	2.3×10^{-2}	2.6×10^{-2}	0.079	+0.106
2800	1.5×10^{-2}	1.6×10^{-2}	0.154	+0.091
5600	6.4×10^{-3}	7.9×10^{-3}	0.236	+0.016
8000	9.1×10^{-3}	9.5×10^{-3}	0.121	−0.011

The strong, significant result is **representation quality**: the relative error of $P_{T_\theta}g$ rises from 0.001 at initialisation to 0.12–0.24 at convergence, $\text{corr}(\log \varepsilon_{\text{equi}}, \text{repr. error}) = -0.835$ ($p = 0.0014$). By the time the network is equivariant, a fifth of the generator direction lies *outside* the resolved tangent space — and that residual is what c_i is computed from.

Secondary readings, with their limits:

- Parameter level, the symmetry-specific margin $r - r'$ averages -0.007 over all checkpoints (0.066 against control 0.074): no signal at any stage of training.
- Unit level it is consistently positive, $+0.085$ (0.491 against 0.407).
- The margin’s decline as equivariance improves is in the predicted direction but **underpowered** at eleven checkpoints ($r = +0.565$, $p = 0.07$). Read as consistent with the mechanism, not established by it.
- $|c_i|$ ’s own stability is flat against equivariance ($p = 0.61$) — as expected of a quantity tracking conditioning rather than symmetry.

6. Scripts

script	establishes
conditioning_decomposition.py	Prop. 1, Cor. 1.1; variance share of conditioning
gauge_dependence.py	Props. 0, 5, 6; cross-gauge overlap; module/unit survival
reparametrisation_covariance.py	Props. 2, 3, 4, 8
attribution_stability.py	identifiability under probe resampling, with a null-vs-null noise floor
probe_grid_sweep.py	whether the subnetwork converges under grid refinement
causal_matched_ablation.py	causal effect at <i>fixed conditioning</i> , via a sensitivity-matched control
compare_runs.py	side-by-side headline table across runs
identifiability_vs_equivariance.py	§5, the regime problem
causal_symmetry_control.py	whether any score causally controls the symmetry, against conditioning-matched baselines

7. Identifiability, measured

Fixing a gauge leaves a separate question: is the top- k set a property of the network or of the probe grid? Resampling 80% of probe rows, 40 draws, with two reference points — $\|S_i\|$ (target-blind, trivially stable, the “easy” level) and r' (null-vs-null, same estimator and same denominator noise as r but no symmetry signal, hence the noise floor).

score	top-20 Jaccard, parameters	top-19 Jaccard, hidden units
$\ S_i\ $ (conditioning reference)	0.776	0.935
$ c_i = J^+g $ (the draft’s score)	0.748	0.881
r_i (conditioning-free)	0.072	0.514
r'_i (null-vs-null control)	0.063	0.397
elastic-net support (gauge-free)	0.179	0.329

$|c_i|$ looks robust and lands within noise of the conditioning reference it is largely made of; the conditioning-free part sits at its own noise floor. At unit granularity r_i separates from its control.

Grid refinement settles the “use more probe points” objection (`probe_grid_sweep.py`, $96 \rightarrow 1792$ rows). The resolved rank is **22 at every resolution** — more probe points add no tangent-space dimensions, so the underdetermination is intrinsic rather than undersampling. $|c_i|$ ’s top-20 does converge toward its finest-grid value ($0.67 \rightarrow 1.00$) — but so does $\|S_i\|$, and $|a_i|$ is flat at 1.00, so that convergence belongs to the conditioning factor. r_i never converges at parameter level (0.00 at every resolution below the finest) while at unit level it reaches 0.65 against 0.09 for its control.

8. Causation, measured

$D = \|\delta\|^2/\|f\|^2$ is zero exactly when the model is $\text{SO}(2)$ -equivariant on the grid and is differentiable in θ , so $\nabla_\theta D$ is the exact per-parameter causal influence on the symmetry.

- **Conditioning predicts it almost completely:** $\rho(\|S_i\|, |\nabla_i D|) = +0.934$. Which parameters causally affect the symmetry is, to first order, just which parameters are sensitive at all.
- **The draft’s own top-20 has no causal effect whatsoever:** perturbing it gives $\Delta \log D = 0.0000$ and $\Delta \log L = 0.0000$, because the column-normalised gauge preferentially selects near-dead parameters ($\rho(|c_i|, \|S_i\|) = -0.88$).
- **After controlling for $\log\|S_i\|$,** the null-calibrated r_i retains a small but significant partial correlation with causal influence: $+0.131$ ($p = 2 \times 10^{-6}$) on the augmented model, $+0.242$ ($p = 0.005$) on the plain one. Small, replicated, and in the right direction.

A caution on the selectivity ratio $\Delta \log D / \Delta \log L$: it is not evidence on its own, because a score selecting near-dead parameters makes both terms ≈ 0 and their ratio noise. `causal_matched_ablation.py` replaces it with the excess over a sensitivity-matched control.